

Dual-ISP Network Assessment and Subnet Remediation

*Multi-Vantage-Point Discovery, Overlapping RFC 1918 Remediation, and Dual-Homed
NAS Reconfiguration Across Two Independent Residential ISP Links*

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Abstract—Two independently provisioned residential Internet services were found to terminate on two physically separate Layer-2 domains that had each been configured, by vendor default, with the identical network address 192.168.0.0/24 and the identical default gateway address 192.168.0.1. A network-attached storage (NAS) appliance was multi-homed across both domains, placing two of its interfaces in overlapping address space. This paper documents the discovery, diagnosis, remediation, and validation of that condition. Purpose-built inventory tooling was developed for Windows PowerShell and macOS Bash to perform ICMP sweep, ARP/neighbor-cache correlation, and limited TCP service probing, and was executed from three vantage points spanning both domains. Because the two domains presented an identical gateway address, differentiation was achieved through gateway hardware-address correlation rather than through Layer-3 evidence. Remediation migrated the second domain to 192.168.20.0/24 and statically readdressed the NAS’s second interface, restoring management and file-sharing services while preserving independent upstream connectivity and a single default route on the multi-homed host. A structured security review and an explicit statement of methodological limitations are included. The work demonstrates that address-plan defects can persist indefinitely without producing an outage, and that single-vantage-point discovery is insufficient to characterize a multi-domain environment.

Index Terms—network assessment, RFC 1918, overlapping subnets, multi-homing, host discovery, address resolution protocol, DHCP, network-attached storage, security hardening.

I. INTRODUCTION

Private IPv4 address space, as allocated by RFC 1918 [1], is reusable by construction: the same prefix may appear in an unbounded number of administratively independent networks without conflict, so long as those networks never need to exchange traffic. Consumer network equipment exploits this property aggressively. The overwhelming majority of shipped residential gateways adopt a default LAN configuration drawn from a very small pool of prefixes, of which 192.168.0.0/24 and 192.168.1.0/24 are the most common. The consequence is that two gateways from the same vendor, or even from different vendors, will frequently arrive preconfigured with byte-identical address plans.

That default is harmless in the single-gateway case for which it was designed. It ceases to be harmless as soon as a premises hosts more than one routed domain. This paper reports on such an environment: a residence provisioned with two independent Internet access services, each terminated on its own gateway, with a four-bay NAS appliance physically cabled into both domains through its two Ethernet interfaces. Both gateways operated on 192.168.0.0/24 with the gateway address 192.168.0.1, and both operated independent DHCP services issuing leases from overlapping ranges.

The condition did not manifest as an outage. Each domain functioned correctly in isolation, and the defect surfaced only as a set of symptoms that are easily misattributed: hosts on one domain could not reach hosts on the other despite apparently sharing a subnet; inventories collected from different machines

disagreed; and the NAS was reachable at two different addresses that were indistinguishable, from Layer 3 alone, as belonging to two different networks. This class of latent defect is precisely the class that is expensive to diagnose under time pressure, and it is the reason the environment was treated as an assessment target rather than as a repair task.

The contributions of this work are as follows.

1. A cross-platform host-discovery toolchain, implemented independently for Windows PowerShell 5.1 and macOS Bash 3.2, producing machine-readable and human-readable inventories from an identical logical pipeline.
2. A multi-vantage-point discovery methodology that establishes domain separation through gateway hardware-address differentiation when Layer-3 evidence is degenerate.
3. A remediation design that eliminates address overlap while preserving two independent upstream paths, dual-domain reachability of a multi-homed host, and a single deterministic default route on that host.
4. An explicit account of the methodology’s false-negative characteristics, which is treated here as a required result rather than as an appendix.

II. BACKGROUND

A. Address Overlap

Overlapping private address space is a well-documented operational hazard in enterprise contexts, where it typically appears after a corporate merger or when a remote-access client attaches to a corporate network from a home network using the same

prefix. The failure mode is consistent: a host resolves a destination address to its own local interface and never forwards the packet, so the destination is unreachable in a manner that produces no error condition upstream. RFC 6890 [2] formalizes the special-purpose registry that governs these ranges. The hazard reported in this paper is the intra-premises variant, which receives comparatively little attention because it is assumed not to arise in a single-gateway residence.

B. Discovery Primitives

Active host discovery on a broadcast domain conventionally combines ICMP echo solicitation with inspection of the local address-resolution cache defined by RFC 826 [3]. The two primitives are complementary and neither is sufficient. ICMP is routinely suppressed by host firewalls, most notably by the default Windows Defender Firewall profile, so a live host may be entirely invisible to a sweep. ARP, by contrast, records any host that has recently participated in Layer-2 resolution and therefore recovers a subset of ICMP-silent hosts, but it is confined to the local domain and decays with the cache timeout. Layer-3 evidence alone cannot distinguish two domains that share a prefix; hardware addresses, which are globally administered from an IEEE-assigned organizationally unique identifier [4], can.

C. Multi-Homing

A host with interfaces in two domains must resolve which interface originates traffic that is not destined for either directly connected prefix. RFC 1122 [5] distinguishes the weak and strong host models that govern this behavior. A multi-homed appliance configured with two default routes of equal preference will select between them nondeterministically, producing intermittent and difficult-to-reproduce failures. The remediation described in Section VII therefore treats default-route determinism as a design requirement rather than as a tuning detail.

III. ENVIRONMENT

A. Topology and Hardware

The assessed environment comprises two consumer gateways, each attached to a separate Internet access service, an unmanaged Ethernet switch on each domain, one NAS appliance cabled into both domains, and a population of wired and wireless client devices. Table I summarizes the infrastructure. Hardware addresses are reported with the organizationally unique identifier retained and the device-specific octets masked; the OUI is sufficient to support every argument made in this paper, and the full address is withheld because a gateway's wireless BSSID is typically adjacent to its wired hardware address and is therefore geolocatable through public wardriving databases. The same masking is applied to the topology figures.

Both NAS interfaces negotiated 1 Gb/s full duplex and exposed an identical service profile: SSH (22), HTTP (80), NetBIOS session service (139), HTTPS (443), SMB (445), DSM management over HTTP and HTTPS (5000 and 5001), and Synology Drive (6690). The consecutive terminal octets of the two

TABLE I
INFRASTRUCTURE INVENTORY

Device	Role	Hardware address
TP-Link Archer AX53 (AX3000)	Gateway A, Service A	3C:6A:D2:XX:XX:XX
TP-Link Archer AX10 (AX1500)	Gateway B, Service B, PPPoE WAN	24:2F:D0:XX:XX:XX
Synology DS423+	NAS, dual-homed	LAN1 90:09:D0:XX:XX:37 LAN2 90:09:D0:XX:XX:38
Unmanaged switches (×2)	Layer-2 distribution	—

TABLE II
DISCOVERY VANTAGE POINTS

Host	Platform	Domain	Address
Workstation W1	Windows 11 Pro, Ryzen 7	A	192.168.0.135
Mac mini M1	macOS 26.3, M4 Pro	A	192.168.0.201
Mac mini M2	macOS 26.5, M2 Pro	B	192.168.0.8

hardware addresses are the primary evidence that the two observed interfaces belong to a single physical chassis rather than to two appliances.

B. Vantage Points

Four general-purpose hosts were available. Three participated in discovery, distributed across both domains, and are enumerated in Table II. The fourth, a Windows 11 workstation, was subjected to a separate read-only host diagnostic and did not appear in any network inventory; the significance of that absence is addressed in Section X.

C. Scope and Authorization

All assessed equipment is owned and administered by the author, and all activity was conducted on premises under the author's control. The tooling described in Section IV performs enumeration only. It does not attempt authentication against any service, does not transmit credentials, does not exercise any vulnerability, and does not alter the configuration of any device it observes. Configuration changes described in Section VII were performed manually and deliberately through vendor management interfaces. The assessment follows the general structure recommended for technical security testing in NIST SP 800-115 [6], restricted to its discovery and review activities.

IV. METHODOLOGY

A. Toolchain Design

Two inventory implementations were developed against a common logical pipeline so that results from heterogeneous operating systems would be directly comparable. The Windows implementation targets Windows PowerShell 5.1 without exter-

nal module dependencies; the macOS implementation targets the Bash 3.2 interpreter shipped with the platform, which constrains it to POSIX-compatible constructs and excludes associative arrays and several parameter expansions available in later shells. Both emit an identical set of artifacts: an HTML report, a PDF rendering, a CSV device inventory, a JSON representation for downstream processing, and a run log.

The pipeline proceeds in five stages:

1. *Interface enumeration.* Local adapters, addresses, prefixes, routes, and configured gateways are read from the operating system. Candidate target prefixes are derived from directly connected RFC 1918 interfaces unless overridden.
2. *Sweep.* Each address in the candidate prefix is probed with a bounded-timeout ICMP echo request, executed concurrently.
3. *Neighbor correlation.* The local address-resolution cache is read after the sweep and merged with the echo results, recovering hosts that suppressed ICMP but responded to Layer-2 resolution.
4. *Service probing.* A fixed set of common TCP ports is tested against each responding address with a short connect timeout.
5. *Classification and rendering.* Hardware-address prefixes, resolved names, and observed service profiles are combined into a provisional device classification, and reports are generated.

Classification at stage 5 is explicitly evidential rather than authoritative. No authenticated fingerprinting is performed, and the tooling does not claim device identity; it reports the observations from which an identity may be inferred. Section V records a case in which that inference was initially wrong.

B. Implementation Defects Encountered

Five defects were identified and corrected in the Windows implementation during development, four of which arose from PowerShell 5.1 semantics rather than from logic errors: an adapter lacking a configured gateway produced a null property dereference; a hexadecimal subnet mask was interpreted as a signed quantity and failed conversion to an unsigned type; asynchronous echo tasks produced a type mismatch on collection; echo results were returned as a nested object whose liveness property could not be resolved directly; and a host with no responding TCP port produced an empty-array parameter fault. The macOS implementation was revised from sequential to concurrent probing, reducing wall-clock runtime for a /24 prefix from approximately eight minutes to approximately 27 seconds.

C. Multi-Vantage-Point Rationale

A single inventory characterizes exactly one broadcast domain. In an environment where domain count is itself unknown, this is a circular constraint: the scan cannot reveal the existence of the domain it is not attached to. Discovery was therefore executed from hosts on both domains, and the resulting inventories were compared as a set rather than consumed individually. The comparison, not any single inventory, constitutes the diagnostic result.

TABLE III
RESPONDING HOSTS BY DOMAIN, PRE-REMEDIATION

Address	Observation
<i>Domain A (from W1 and M1)</i>	
192.168.0.1	Gateway A; DNS, HTTP, HTTPS
192.168.0.2	NAS interface LAN1
192.168.0.45	Unidentified; locally administered address
192.168.0.118	Unidentified; persistent across both scans
192.168.0.135	Workstation W1
192.168.0.156	Unidentified; locally administered address
192.168.0.201	Mac mini M1; TCP 5000 open
<i>Domain B (from M2)</i>	
192.168.0.1	Gateway B; distinct hardware address
192.168.0.8	Mac mini M2
192.168.0.52	Unidentified
192.168.0.56	Unidentified; ARP-visible, ICMP-silent
192.168.0.107	Unidentified
192.168.0.228	Unidentified; TCP 80 open
192.168.0.242	NAS interface LAN2

V. RESULTS

A. Divergent Inventories

The three inventories reported materially different device populations while reporting an identical prefix and an identical default gateway address. Table III consolidates the responding addresses, and Fig. 1 shows the consolidated Domain A view produced before the second domain was known to exist.

B. Gateway Hardware-Address Differentiation

The decisive observation is that the two domains reported the same gateway address with different hardware addresses:

Vantage M1 (Domain A)	Vantage M2 (Domain B)
gateway 192.168.0.1	gateway 192.168.0.1
hwaddr 3C:6A:D2:XX:XX:XX	hwaddr 24:2F:D0:XX:XX:XX

Two distinct globally administered hardware addresses cannot belong to one interface. The two vantage points were therefore resolving the same Layer-3 address to two different physical devices, which is possible only if they occupy separate Layer-2 domains. This established domain separation without requiring any change to the environment, and it did so using evidence that a Layer-3-only methodology would have discarded.

C. Identification of the Multi-Homed Host

Two addresses, 192.168.0.2 on Domain A and 192.168.0.242 on Domain B, presented identical service profiles and hardware addresses differing only in the terminal octet. The natural reading, that these are two network interface controllers on one chassis assigned consecutively at manufacture, was subsequently confirmed through the appliance's own management interface. The NAS was therefore simultaneously present in both overlapping domains.

D. A Falsified Hypothesis

The initial single-vantage inventory classified 192.168.0.201 as a probable storage appliance on the evidence of an open TCP port 5000, which is the appliance vendor's unencrypted manage-

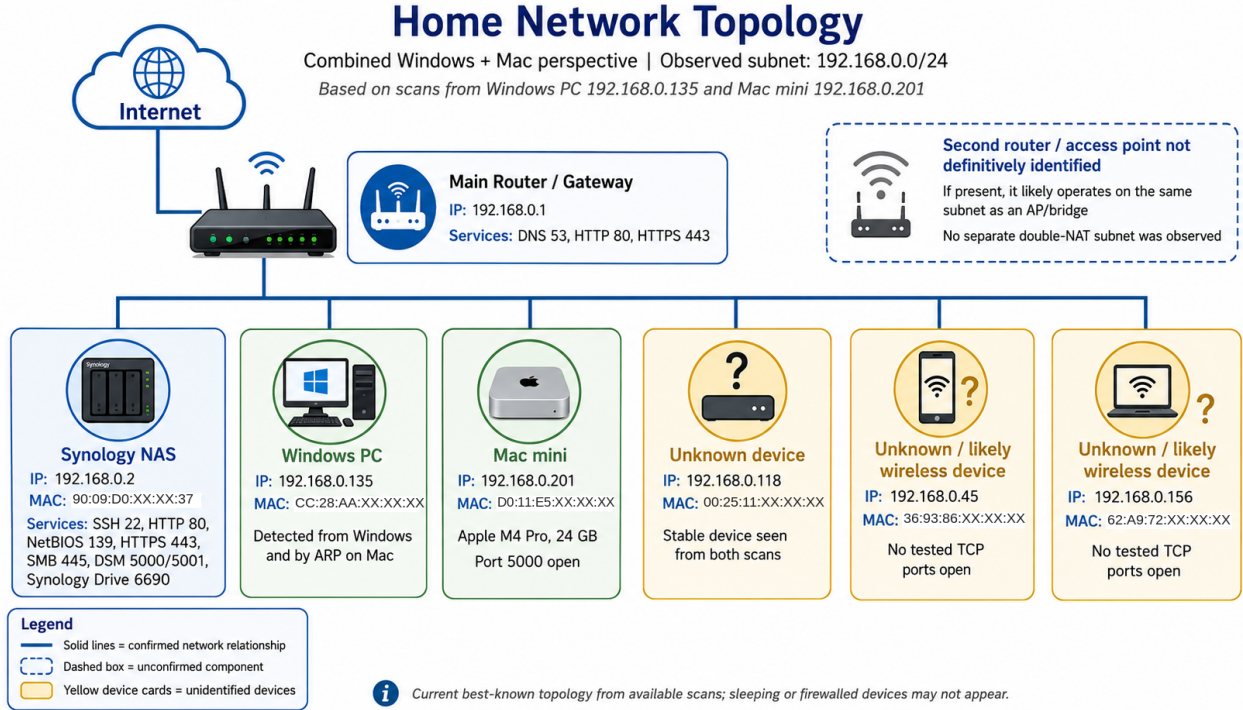


Fig. 1. Consolidated view of Domain A from the Windows and macOS vantage points. At this stage the environment was still assumed to contain a single gateway, and the second domain was not yet known to exist. Hardware addresses are masked to the organizationally unique identifier.

ment port. The subsequent inventory executed from that host established it as a Mac mini running an unrelated service on the same port. The hypothesis was formed from a single weak signal and was retracted when a second vantage point contradicted it. It is retained in this report because it is a concrete demonstration of why service-profile inference must not be promoted to device identity, and because the correction is itself a result of the multi-vantage-point methodology.

VI. DIAGNOSIS

The environment contained two physically separate Layer-2 domains that were logically indistinguishable, as summarized in Fig. 2. The consequences, in order of operational severity, were as follows.

Ambiguous management plane. Both gateways answered at 192.168.0.1. A management session established from an arbitrary host reached whichever gateway that host happened to be attached to, with no indication in the interface itself of which device was being configured. Any configuration change carried an unmitigated risk of being applied to the wrong device.

Address collision under DHCP. Two independent DHCP servers [7] issued leases from overlapping ranges within the same prefix. A given address could be validly assigned on both domains simultaneously. No conflict was detected, because conflict detection is scoped to a broadcast domain and the two servers could not observe each other.

Non-deterministic behavior on the multi-homed host. The NAS held interfaces in two domains that it could not distinguish

by prefix. Any routing decision that was not to a directly connected address had no unambiguous basis for interface selection.

Degraded diagnosability. Incomplete and mutually inconsistent inventories were the outward symptom. Absent hardware-address correlation, the most available explanation for the symptom set is client isolation or a firewall policy, both of which would have led troubleshooting away from the actual defect.

VII. REMEDIATION

A. Design Constraints

Three constraints shaped the design. First, both upstream services were to remain independent and in use; consolidating onto one service was out of scope. Second, the NAS was to remain reachable from both domains at native line rate, which excluded routing NAS traffic across a gateway. Third, no transit path was to be created between the two domains, since each carries its own untrusted upstream and merging their broadcast domains would defeat the isolation the topology already provides.

Several candidate designs were rejected against these constraints. Bridging the two domains through the NAS would have created exactly the transit path constraint three prohibits, and would additionally place a storage appliance in a forwarding role for which it is not designed. Link aggregation across the two NAS interfaces is invalid because the interfaces terminate on different switches in different domains, and aggregation requires a single logical peer. Disabling DHCP on one gateway, a common remedy where a second router is deployed as an access point, is inapplicable here because each gateway serves its own

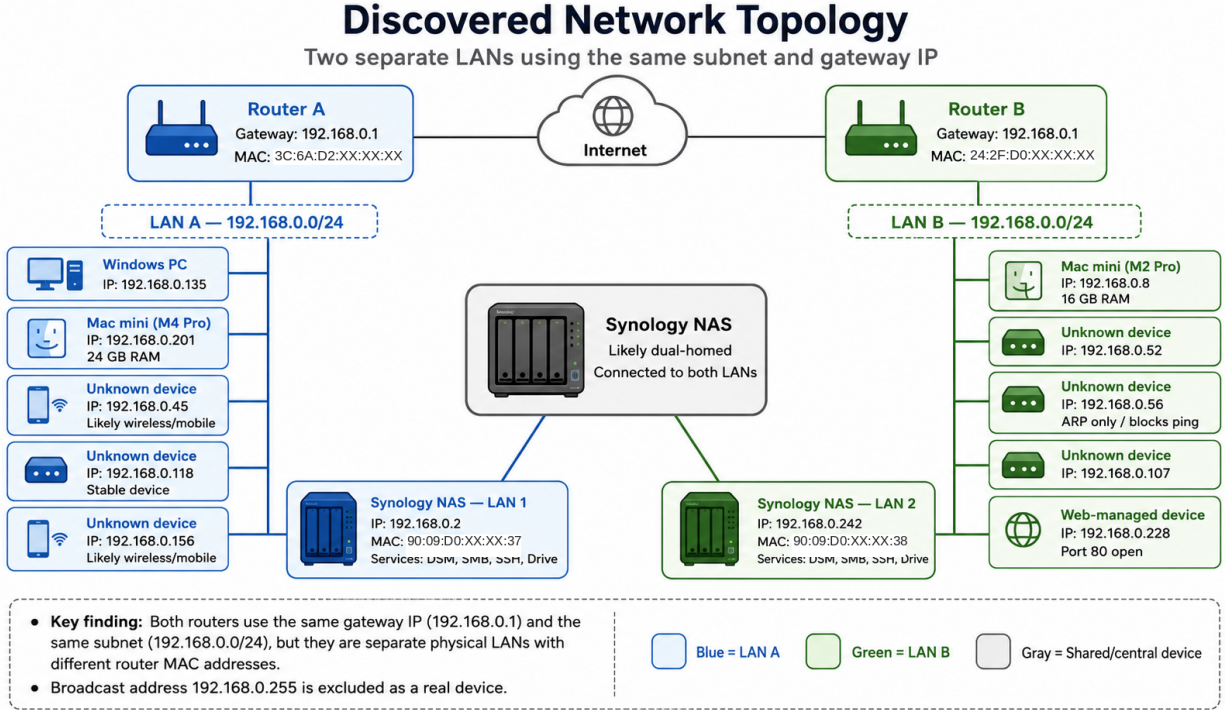


Fig. 2. Diagnosed topology. Two physically separate Layer-2 domains present the identical prefix 192.168.0.0/24 and the identical gateway address 192.168.0.1, and are distinguishable only by the gateway hardware address. The NAS is multi-homed across both. The directed broadcast address 192.168.0.255 is annotated as a scan artifact rather than a host.

independent upstream and must issue its own leases.

B. Implemented Change

The retained design assigns disjoint prefixes to the two domains and leaves the topology otherwise unaltered. The resulting design is shown in Fig. 3. Gateway B was migrated from 192.168.0.1/24 to 192.168.20.1/24 with its DHCP scope relocated accordingly. Gateway A was left unchanged. The NAS interface on Domain B was converted from its prior address to a static assignment of 192.168.20.2/24, and was deliberately not configured with a default gateway, leaving the interface on Domain A as the appliance’s single default route. Upstream configuration on Gateway B, including PPPoE credentials and NAT behavior, was not modified.

C. Transient Loss of Service

Immediately following the gateway migration, the client attached to Domain B lost connectivity. The cause was a stale DHCP lease: the client continued to assert an address in the retired prefix while the gateway had moved to the new one, placing the two on different logical networks across a functioning physical link. This is an expected consequence of changing a served prefix without first shortening the lease time, and it resolved on forced lease renewal. It is reported here because the failure is indistinguishable at the client from a hardware fault, and because the sequencing lesson, that lease duration should be reduced in advance of a planned readdressing, generalizes directly to production change windows.

D. Service Restoration

The NAS management interface was reached over the still-functional Domain A path while the Domain B interface was reconfigured, which avoided any requirement for console access. Following the change, the appliance’s discovery utility reported the second interface as statically addressed at 192.168.20.2 and operational. Client mounts on Domain B were repointed from the retired address to the new one.

VIII. VALIDATION

Validation covered reachability, service availability, and the absence of unintended transit. Each domain’s gateway and NAS interface responded to echo requests from within that domain. Management access to the NAS was confirmed over HTTPS on port 5001 from both domains, and SMB mounts were re-established from both domains. Upstream reachability was confirmed independently on each service. DHCP was confirmed to be issuing leases from the correct scope on each gateway.

A negative test was included deliberately: hosts on Domain A remain unable to reach hosts on Domain B, and the converse. This is the intended outcome. The remediation resolves address ambiguity; it does not, and should not, establish inter-domain connectivity. Conflating the two objectives is a common error in this scenario, and verifying the absence of transit is as much a part of validation as verifying reachability.

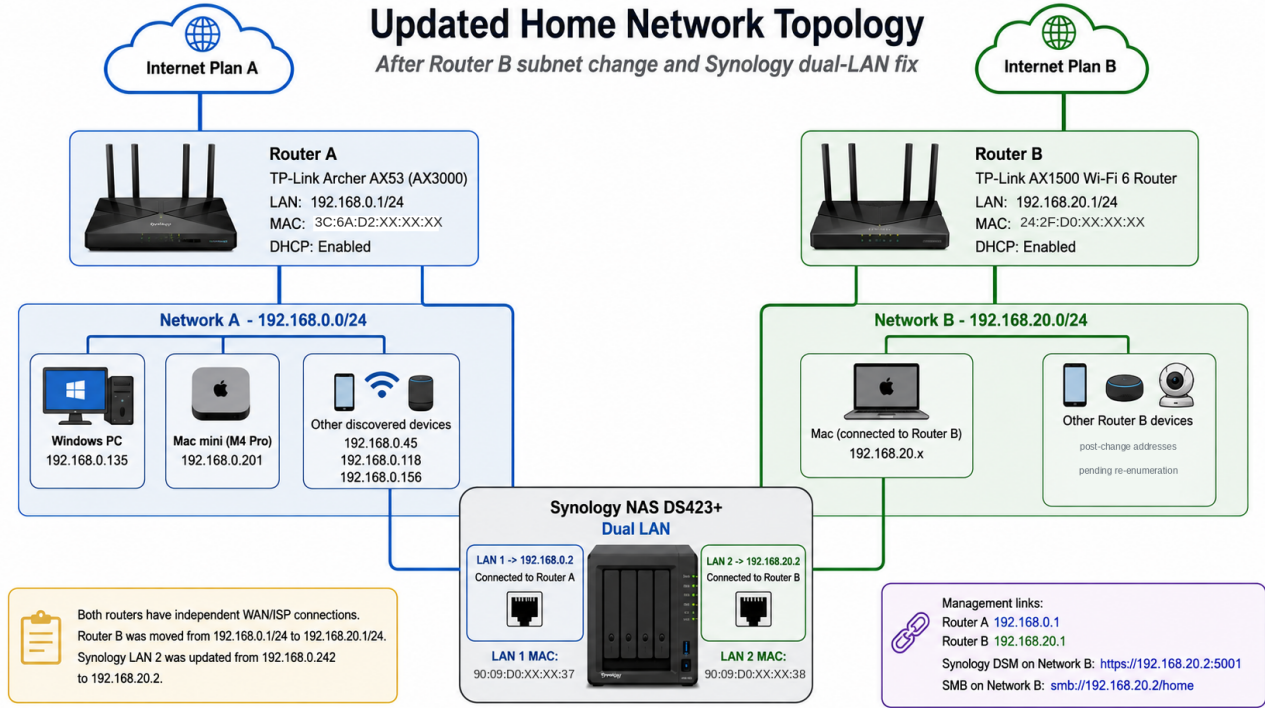


Fig. 3. Remediated topology. Domain B is migrated to 192.168.20.0/24 and the NAS interface on that domain is statically readdressed to 192.168.20.2, while Domain A is unchanged. Each domain retains an independent upstream service and no transit path is created between them. Domain B client addresses were reassigned by DHCP from the new scope and are shown as pending re-enumeration rather than carried over from their pre-change values.

IX. SECURITY REVIEW

A structured review was conducted concurrently with remediation. Table IV records the findings and their dispositions. Ratings are qualitative and reflect the residential threat model, in which Internet-facing exposure and lateral movement from compromised consumer-grade endpoints dominate the risk profile.

The unidentified-host finding is rated highest not because any specific device is known to be malicious, but because an environment in which the administrator cannot enumerate what is attached cannot support any meaningful access-control assertion. Two of the unidentified hosts presented locally administered hardware addresses, consistent with the address-randomization behavior now default in mobile operating systems [8], which reduces the diagnostic value of hardware-address correlation for that class of device and requires resolution through the gateway's own client table instead.

X. LIMITATIONS

The methodology has known false-negative characteristics, enumerated here because an inventory presented without them invites overinterpretation.

ICMP suppression. Hosts configured to discard echo requests are invisible to the sweep and are recovered only if they appear in the address-resolution cache. The fourth workstation in the environment produced a complete host diagnostic yet appeared in no inventory. The available explanations, that it was powered down, that its firewall suppressed both echo and every probed

port, or that it is attached to a domain not scanned, are not distinguished by the collected evidence, and the case is reported as unresolved.

Point-in-time sampling. Each inventory is a single observation. Devices that were asleep, disconnected, or on battery-conserving wireless states during the sweep are absent by construction. No longitudinal collection was performed.

Address randomization. Locally administered hardware addresses defeat vendor attribution and are not stable across associations, so a randomizing device cannot be tracked across inventories by hardware address.

Reporting artifacts. The macOS implementation counted multicast and broadcast destinations, specifically 224.0.0.251, 239.255.102.18, and the directed broadcast address, among its responding hosts, inflating the reported device total. These are not hosts. The defect is in the reporting filter rather than in the discovery logic, but it affects every published count from that implementation and is corrected in the current revision.

Evidential classification. Device type is inferred from hardware-address prefix, resolved name, and observed service profile. None of these is authoritative, and Section V-D documents a case in which the inference was incorrect.

Post-change enumeration. The inventories reported here pre-date remediation. Client addresses on Domain B were reassigned by DHCP from the new scope and are not derivable from their prior values; a post-change collection is required before the Domain B client census can be restated.

TABLE IV
SECURITY FINDINGS

Finding	Rating	Disposition
Unidentified hosts on both domains	High	Correlate against gateway client lists; deny unaccounted devices
Unencrypted management port 5000 reachable on NAS	Med	Restrict to HTTPS 5001; disable plaintext listener
No host-level firewall policy on NAS	Med	Permit only 192.168.0.0/24 and 192.168.20.0/24
Single-factor administrative access to NAS	Med	Enable second factor; disable default administrative account
Firmware currency not verified on either gateway	Med	Establish update cadence
Wireless credentials exposed in working photographs	Med	Rotate; verify WPA2/WPA3-AES, disable WPS
No segmentation for IoT and guest endpoints	Med	Isolate on guest SSID; deny access to storage
Legacy SMB dialect not explicitly disabled	Low	Enforce SMB2/SMB3 minimum
Backup and snapshot strategy undefined	Med	Define retention; maintain off-site copy
Physical installation obstructs appliance airflow; cabling unlabeled	Low	Reorganize; label both ends of every run

XI. DISCUSSION AND FUTURE WORK

The general lesson is that identical private addresses observed from different vantage points carry no implication of identical devices, and that the reverse inference, treating a shared prefix as evidence of a shared domain, is unsound in any environment where more than one gateway may exist. The corollary is methodological: where Layer-3 evidence is degenerate, Layer-2 evidence remains discriminating, and a discovery tool that discards hardware-address data forfeits its ability to detect this entire class of defect.

A second observation concerns the relationship between defect severity and defect visibility. The condition described here persisted indefinitely without producing an outage, because each domain was internally consistent. Latency between introduction and detection is characteristic of address-plan defects generally, and it is an argument for periodic assessment of environments that are believed to be functioning correctly.

Planned extensions, in order of value, are: a post-remediation collection on both domains with a structured differential against the pre-remediation baseline; resolution of every unidentified host against gateway client tables; static reservations for infrastructure addresses so that the address plan is documented in the DHCP configuration itself; introduction of a managed switch to permit VLAN segmentation of untrusted endpoints, which is the principal capability the current unmanaged Layer-2 distribution cannot provide; centralized log collection from both gateways and the appliance; and longitudinal availability monitoring to convert the point-in-time inventory into a continuous one.

XII. CONCLUSION

Two independently provisioned residential networks were found to occupy identical RFC 1918 address space with identical gateway addresses, with a storage appliance multi-homed across both. The condition was detected through multi-vantage-point discovery and diagnosed through gateway hardware-address correlation, an approach that succeeded specifically because Layer-3 evidence was degenerate. Remediation assigned disjoint prefixes, statically readdressed the appliance’s second interface, and preserved a single deterministic default route, restoring management and file-sharing services on both domains without creating transit between them. The work is reported together with its false-negative characteristics and one falsified classification hypothesis, on the view that an assessment that does not state its own blind spots overstates its coverage.

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